Exact block and orbit deficits at a single crossing

Human Compression project: bounded L044 proof component

Let $G = \langle \rho_0, \dots, \rho_{r-1} \rangle \le A_n$ be transitive, with every maximal omission subgroup G_j intransitive. Suppose G_i has two orbits A_1, A_2, and its action on A_j preserves $m_j \ge 2$ blocks of size $k_j \ge 2$. Thus $n = k_1 m_1 + k_2 m_2$. In the single-crossing application these are the two orbits joined by the unique eligible i-edge. The counting argument below does not require the string property or IP.

Choose a minimal subset M of the distinguished generators of G_i whose images generate the complete simultaneous action on the two block systems. Let c_j be the number of $\langle M \rangle$-orbits on A_j, and put $c = c_1 + c_2$. Minimality makes the block images of M independent; it does not assert that their omission actions are intransitive or that their projected marking has IP.

Lemma 1 (A forest on actual subgroup orbits)**.** *There is a forest on the c orbits of $\langle M \rangle$, with precisely one edge of each distinguished colour not in M. In particular*

$$|M| \le m_1 + m_2 - 2, \qquad c_j \le k_j, \qquad r - |M| \le c - 1.$$

Proof. The first bound is the retained intransitive independent-length bound for a subgroup of $S_{m_1} \times S_{m_2}$, obtained by successive orbit projections and the normal-quotient normalization lemma together with Whiston's bound. This statement does not require an intersection property for the quotient.

The group $\langle M \rangle$ is transitive on the m_j blocks of A_j. Each of its point orbits therefore meets every block, so has at least m_j points. This proves $c_j \le k_j$.

For every colour l not in M, choose an actual ρ_l-edge joining different G_l-orbits. Such an edge exists: otherwise ρ_l preserves every G_l-orbit setwise and adjoining it cannot make the whole group transitive. Contract the point orbits of $\langle M \rangle$. Both ends of the chosen edge lie in different contracted vertices, since $\langle M \rangle \le G_l$.

The chosen edges form a forest. For if a cycle existed, removing its l-edge would leave a path between its endpoints made of other-colour edges and paths inside $\langle M \rangle$-orbits. All these steps are performed by elements of G_l, contradicting the choice of the l-edge. The same argument excludes parallel edges as cycles of length two. A forest on c vertices has at most $c - 1$ edges, proving the last inequality. □

Define the nonnegative integer deficits

$$\alpha = m_1 + m_2 - 2 - |M|, \quad \beta = k_1 + k_2 - c, \quad \gamma = c - 1 - (r - |M|),$$

and put $D = (k_1 - 2)(m_1 - 2) + (k_2 - 2)(m_2 - 2)$.

Lemma 2 (The exact rank deficit)**.** *One has*

$$n - 1 - 2r = D - 3 + 2(\alpha + \beta + \gamma).$$

Consequently the desired bound $r \le \lfloor (n-1)/2 \rfloor$ holds whenever $D + 2(\alpha + \beta + \gamma) \ge 3$.

Proof. The definitions give $r = k_1 + k_2 + m_1 + m_2 - 3 - \alpha - \beta - \gamma$. Substitute this into $n - 1 - 2r$ and expand D. □

Lemma 3 (Saturation forces even block sizes)**.** *If $\alpha = \beta = 0$, both k_1 and k_2 are even.*

Proof. The block image has an independent family of size $m_1 + m_2 - 2$ and exactly two orbits. The accepted intransitive equality lemma from L047 therefore identifies that image as the full product $S_{m_1} \times S_{m_2}$.

The equality $c_j = k_j$ forces every $\langle M \rangle$-orbit in A_j to have size exactly m_j, meeting each block in a single point. On each such orbit, projection to the block system is an equivariant bijection. Thus the action of $\langle M \rangle$ is the diagonal action of its block image on k_j copies of the natural m_j-point action. The union of these orbits is the full set of points, so the block projection is faithful on $\langle M \rangle$.

In the full block image choose an element which is a transposition on the first block system and is the identity on the second. Its unique lift to $\langle M \rangle$ has sign $(-1)^{k_1}$. Since every element of G is even, k_1 is even. Interchanging the systems proves the claim for k_2. □

Lemma 4 (Exact residual numerical configurations)**.** *If the target rank bound fails, either*

1. *$D = 0$ and $\alpha + \beta + \gamma \leq 1$; each orbit's block system has either two blocks or blocks of size two; or*

2. *$D = 2$, $\alpha = \beta = \gamma = 0$, and, up to interchange, $(k_1, m_1) = (4, 3)$ while the other block system has $\min(k_2, m_2) = 2$ and even k_2.*

Proof. Lemma 2 restricts a failure to $D \leq 2$. For $D = 0$ it gives the first case. If $D = 1$ or 2, all three deficits must vanish. A summand equal to one requires $(k_j, m_j) = (3, 3)$, excluded by Lemma 3. Thus neither $D = 1$ nor a sum of two ones is possible. A summand equal to two requires $(k_j, m_j) = (3, 4)$ or $(4, 3)$; the first is again excluded by parity. The other summand is zero, giving precisely case two. □

Lemma 5 (Excluding the broad saturated $D = 2$ family)**.** *Restore the string C-group and unique-crossing hypotheses. The residual case $(k_1, m_1) = (4, 3)$, $(k_2, m_2) = (2, m)$ with $m \geq 3$ and $\alpha = \beta = \gamma = 0$ is impossible.*

Proof. Saturation identifies $\langle M \rangle$ faithfully with $S_3 \times S_m$ on four copies of its first natural action and two copies of its second. Each primitive factor's active indices form an interval next to i by Lemma 7 below. These intervals cannot be on the same side: the two-anchor deficit of Lemma 10 would give $|M| \leq m - 1$, whereas saturation gives $|M| = m + 1$. Thus they lie on opposite sides, and each interval acts trivially on the other block orbit. They end and start, respectively, at the two neighbors of i. This argument includes $m = 3, 4$ and does not require an extra small-marking census.

There are four distinguished generators outside M and i: indeed $c = 6$ and $\gamma = 0$ give $r - |M| = 5$. Only the two outer neighbors of the component intervals can fail to commute with $\langle M \rangle$. At least two of these four generators therefore centralize $\langle M \rangle$. They fix both crossing endpoints, because the only indices that can move them other than i are already in M. Centralizing $\langle M \rangle$ then forces fixation of the whole copy containing each endpoint.

The full centralizer on the remaining copies is S_3 permuting the three unused three-point copies, times the trivial permutation group on the single unused m-point copy. Here natural symmetric actions of degree at least three have trivial internal centralizer; the two factors' different fixed-point sets also prevent exchange of copies between the two original orbits. A transposition of three-point copies is odd on the original points. Hence the centralizer inside A_n is A_3, which has no nonidentity involutions. It cannot contain either of the asserted commuting distinguished generators. This contradiction proves the claim. □

Lemma 6 (An anchored maximal symmetric string)**.** *Let $m \geq 3$, and let $(s_0, \ldots, s_{m-2})$ be an independent involutory string generating S_m. Suppose a point v is fixed by every s_j except s_0. Then the noncommuting graph is the path in this order, and the subgroup $T = \langle s_0, \ldots, s_{m-3} \rangle$ has an orbit containing v of size at least $m - 1$. Moreover $C_{S_m}(T) \cap (S_m)_v = 1$.*

Proof. For $m \geq 5$ use the accepted maximal independent-string classification from L047, including its degree-five and degree-six endpoints. The abstract Coxeter path is ordinary in the natural action, except possibly for the outer degree-six marking. The latter's distinguished involutions are all fixed-point-free, and so it cannot satisfy the hypothesis on v. In the ordinary path v is its first terminal point. The group T is S_{m-1} on the first $m-1$ points, fixing the final point. Its centralizer fixing v is trivial.

For $m = 3$, s_1 is the transposition on the other two points, and s_0 is a transposition joining v to one of them. The assertions follow directly. For $m = 4$, the independent pair s_1, s_2 lies in $(S_4)_v \cong S_3$; it consists of two distinct transpositions. Relabel so that $v = 1$, $s_1 = (23)$, and $s_2 = (34)$. Since s_0 commutes with (34) and moves 1, it is (12) or (12)(34). These give, respectively, $T = S_{\{1,2,3\}}$ and the transitive dihedral group of order eight on four points. In the second case a centralizer element fixing one point of a transitive action is the identity; the first case is immediate as well. Both displayed strings have the asserted noncommuting path. □

Lemma 7 (Primitive block intervals)**.** *Restore the string and unique-crossing hypotheses. If a primitive block image of $\langle M \rangle$ has degree at least three, its active indices in M form one interval, beginning or ending next to i.*

Proof. Partition a disconnected active set at a gap. The two resulting nontrivial commuting normal factors of its primitive image are transitive. A nonidentity element of either factor, since it centralizes the other transitive factor, is fixed-point-free. Every active distinguished generator would therefore move the block containing the crossing endpoint. Only $i-1$ and $i+1$ can do that: every other generator fixes that endpoint itself. There would be exactly two active generators, one in each factor. Each factor would be a transitive group generated by a single involution, forcing degree two, a contradiction. Thus the active set is an interval. Transitivity forces one of its generators to move the crossing block; the interval avoids i and must contain one of its neighbors. □

Lemma 8 (The maximal block-image two-copy branch)**.** *Restore the string C-group and unique-crossing hypotheses. Suppose $(k_1, m_1) = (2, a)$ and $(k_2, m_2) = (2, b)$, where $a, b \geq 3$, and $\alpha = 0$. Then $r \leq a + b - 1 = (n-2)/2$, so the desired rank bound holds.*

Proof. The block image is the full $S_a \times S_b$, with a maximal independent involutory string marking; its lift may have a nontrivial kernel. By Lemma 7, each factor's active indices form an interval ending or starting next to i.

They cannot both lie on the same side. Suppose they did, on the right. Choose a minimal subfamily $L \subset M$ generating the S_b image. Normal-quotient normalization and Whiston's bound force $|L| = b-1$, leaving $a-1$ independent generators outside L. The image of L satisfies Lemma 6, with the crossing block as v. Its connected noncommuting path forces L to occupy the initial interval of indices starting at $i+1$. Every generator in $M \setminus L$ therefore centralizes its far-end omission subgroup, or all of L, and fixes the crossing block. Lemma 6 makes its S_b image trivial. Thus these $a-1$ generators form an independent family in S_a, so generate S_a by Whiston's equality clause. But each fixes the first orbit's crossing block, because its index is not $i+1$. This is impossible. The left-sided case is identical.

The active intervals instead lie on opposite sides and are disjoint. Each acts trivially on the other block orbit, and independence and the full total length force their lengths to be $a-1$ and $b-1$. Both images satisfy Lemma 6, with their near endpoint next to i.

Every distinguished generator x outside M and i fixes both crossing endpoints. It centralizes either both full intervals, or one full interval and the far-end omission of the other. A centralizer element of a transitive permutation group fixing one point is trivial; together with Lemma 6, this makes both block images of x trivial. It fixes each crossing block pointwise, since that block has size two. Commutation transports this fixation throughout the block orbit of the corresponding centralized subgroup. Both original orbits are fixed pointwise, except possibly the single far

two-point block on the side where only the omission subgroup is centralized. Thus x is the identity or a single transposition, both impossible for a distinguished nonidentity element of A_n.

There is consequently no generator outside M and i, so $r = |M|+1 = a+b-1$. No existence assertion is made for this lower-rank configuration. No projected IP or triviality of the lift's kernel has been assumed. □

Lemma 9 (Opposite primitive block intervals)**.** *Restore the string C-group and unique-crossing hypotheses. Suppose each orbit has a system of two-point blocks, with numbers $a, b \geq 3$, and the two induced block actions are primitive. If their active intervals in M lie on opposite sides of i, then $r \leq \lfloor (n-1)/2 \rfloor$.*

Proof. Here $D = 0$. If $\alpha \geq 2$ the exact deficit proves the result, while $\alpha = 0$ is Lemma 8. A failure with $\alpha = 1$ therefore has $\beta = \gamma = 0$. In particular $c = 4$, and precisely two distinguished generators lie outside M and i. Both neighbors of i belong to M.

A distinguished generator x outside M and i cannot centralize all of $\langle M \rangle$. It fixes both crossing endpoints. In each block system its image centralizes the transitive block image of M and fixes the crossing block, so its block image is trivial. It consequently fixes that two-point block pointwise. Commutation transports this fixation to all blocks, making $x = 1$.

The two active intervals are disjoint, each acting trivially on the other's block orbit. Their independent image markings have lengths $u \leq a-1$, $v \leq b-1$. Since $u + v = |M| = a + b - 3$, one attains its maximum, say $u = a-1$. Its image is S_a by Whiston, with the anchored marking of Lemma 6.

The two outside generators must occupy the two outer neighboring indices of the intervals; any generator further away would centralize all of M. Let x be the neighbor on the maximal S_a side. On the other orbit it centralizes the full transitive block image and fixes the crossing block, hence has trivial block image and fixes every point by transporting fixation of the two-point crossing block. On its own side it centralizes the far-end omission subgroup and fixes the crossing block. Lemma 6 makes its block image trivial and propagates pointwise fixation to at least $a-1$ blocks. Thus it is the identity or a single odd transposition, a contradiction. □

Lemma 10 (Two anchored orbits consume two deficits)**.** *Let M be an independent involutory string in $S_a \times S_b$, where $a, b \geq 3$, with both projections transitive. Suppose M occupies one interval and only its first generator t can move specified points v, w in the two images. Then $|M| \leq a + b - 4$.*

Proof. Transitivity forces t to move both v and w. Every generator other than t and its neighbor fixes v, w and commutes with t, so also fixes the distinct points v^t, w^t. Those $|M| - 2$ generators remain independent in the simultaneous image, and lie in $S_{a-2} \times S_{b-2}$. Normal-quotient normalization and Whiston's bound give

$$|M| - 2 \leq (a-3) + (b-3).$$

This includes $a = 3$ or $b = 3$, when the corresponding remaining one-point factor contributes zero. The asserted inequality follows. A one-generator string cannot be transitive in degree at least three, so the displayed removal of two generators causes no missing endpoint case. □

Lemma 11 (The complete two-point-block branch)**.** *Restore the string C-group and unique-crossing hypotheses. Suppose both orbit block systems have size two, have primitive block images, and have at least three blocks. Then $r \leq \lfloor (n-1)/2 \rfloor$.*

Proof. Both active sets in M are intervals next to i by Lemma 7. If they are on opposite sides, use Lemma 9. If they are on the same side, their union is a single interval. Only its generator next to i can move either crossing block. Lemma 10 gives $\alpha \geq 2$, and the exact deficit, with $D = 0$, proves the result. □

Lemma 12 (A binary anchored string)**.** *Let $m \geq 4$. Suppose $(t_0, \ldots, t_{m-2})$ is an independent involutory string in $C_2 \times S_m$ whose second projection is transitive. Suppose only t_0 has nonidentity first projection, and only t_0 can move a specified point v of the second projection. Put $T = \langle t_0, \ldots, t_{m-3} \rangle$ in the second projection. Its orbit containing v has size at least $m-1$, and $C_{S_m}(T) \cap (S_m)_v = 1$.*

Proof. The final $m-2$ generators have trivial first projection and fix v. Their second projections remain independent. Whiston therefore makes them generate the full S_{m-1} on the other points. For $m-1 \geq 5$ their maximal independent-string marking is an abstract Coxeter path by L047. An outer degree-six marking is impossible here: its endpoint omission is transitive on all six points, so a permutation centralizing it must fix its unique fixed point v. The first generator, which centralizes this omission and moves v, rules this out. In the ordinary marking, the omission of its first generator is S_{m-2}, fixing v and one more point u. Its centralizer is generated by (vu), so the second projection of t_0 is (vu). The complete projected marking is the ordinary anchored Coxeter path.

For $m = 4$, the two tail generators are two distinct transpositions in S_3. The direct calculation in Lemma 6 gives exactly the same two possibilities there: the first second-projection generator is (12) or $(12)(34)$, followed by $(23), (34)$. Their initial omission subgroups are S_3 on three points or transitive D_8.

It remains to check $m = 5$, where the tail is a maximal independent involutory string of S_4. Its two commuting end involutions cannot both be double transpositions, since then they lie in the normal four-group and adjoining one involution cannot generate S_4. If they are disjoint single transpositions, the middle generator must be a transposition joining their pairs: a double transposition would only generate a group preserving the partition into those pairs. This is the ordinary Coxeter path. The remaining case, up to reversal and relabeling, is

$$((23)(45), (34), (45))$$

on the four points other than $v = 1$. Indeed a commuting single and double transposition share one factor, and the middle involution must be a single transposition connecting their two pairs; a double transposition again preserves that pair partition. If the displayed tail is reversed, its first omission is transitive D_8, making it impossible for t_0 to commute with that omission and move 1.

For the displayed order, t_0 has second projection (12). This order contradicts independence in $C_2 \times S_5$. Write ϵ for the nonidentity element of C_2. The first three generators are

$$(\epsilon, (12)), \quad (1, (23)(45)), \quad (1, (34)).$$

Successive conjugation of the first by the second, third and second yields $(\epsilon, (13))$, $(\epsilon, (14))$ and $(\epsilon, (15))$. These lifted star transpositions generate the graph of the sign homomorphism $S_5 \to C_2$. The third generator $(1, (34))$ lies outside that graph, so the first three already generate $C_2 \times S_5$. The final $(1, (45))$ would be redundant. Thus only the ordinary tail survives at $m = 5$. In all surviving cases, the stated orbit and centralizer conclusions follow from the explicit paths or D_8. □

Lemma 13 (A four-point orbit in the two-point-block branch)**.** *Restore the string C-group and unique-crossing hypotheses. Suppose the two orbits have systems of two-point blocks, with primitive block images of degrees 2 and $b \geq 4$. Then $r \leq \lfloor (n-1)/2 \rfloor$.*

Proof. Here $n = 4 + 2b$ and $D = 0$. Only $\alpha = 0, 1$ can violate the bound. The degree-b active set in M is an interval by Lemma 7; assume it starts at $i+1$. The only possible active indices on the two-block orbit are $i-1, i+1$, since any block exchange moves its crossing endpoint's block.

Suppose first $i-1 \in M$. Its block image is the nonidentity C_2 element and is trivial on the other orbit. The rest of M is the degree-b interval. Both neighbors of i are in M, so all outside generators fix both crossing endpoints. An outside generator centralizing M has trivial block images and fixes the two crossing blocks pointwise; commutation transports that fixation to every block, a contradiction. Consequently outside generators can occur only at the two outer neighbors of M.

If $\alpha = 1$, failure forces $\beta = \gamma = 0$, so exactly two such generators occur. Consider the outer neighbor on the single-generator side. It centralizes the complete transitive degree-b block image and fixes its crossing endpoint, hence fixes that whole original orbit pointwise. On the four-point orbit it fixes the crossing block setwise (a permutation of two blocks fixing one is trivial), and fixes that block pointwise. Only the other two-point block remains, so the generator is the identity or one odd transposition, impossible.

If $\alpha = 0$, the other interval has $b-1$ generators. Its degree-b images are independent: any redundancy there could be lifted to redundancy in M using the pure C_2 generator. They consequently form the anchored maximal S_b marking of Lemma 6. The outer neighbor on the single-generator side is excluded as above. At the other outer neighbor, commutation with the pure C_2 generator fixes the four-point orbit pointwise, by transporting its fixed crossing block. Lemma 6 fixes every point on the other orbit except possibly its far two-point block. Parity again excludes this neighbor. Thus there is no outside generator; the rank is $|M| + 1 = b + 1$, already below the claimed upper bound.

Now suppose $i-1 \notin M$. Then M is the single right-hand interval and only its first generator can exchange the two blocks in the four-point orbit or move the other crossing block. Removing its first two generators leaves an independent family fixing both degree-two blocks and two degree-b blocks, of length at most $b-3$. Thus $|M| \leq b-1$, excluding $\alpha = 0$. With $\alpha = 1$, failure again forces $\beta = \gamma = 0$ and two outside generators. Apart from $i-1$ and the far outer neighbor, any outside generator fixes both crossing endpoints and centralizes M, which is impossible as above. Hence the far outer neighbor x occurs.

Apply Lemma 12 to the simultaneous block marking M. The generator x fixes both crossing endpoints. On the four-point orbit its block action is trivial, since it fixes one of two blocks; it centralizes the first generator of M, which exchanges those blocks, and therefore fixes the whole orbit pointwise. On the other orbit it centralizes the far-end omission subgroup of M. Lemma 12 makes its block action trivial and propagates fixation from the crossing block to at least $b-1$ blocks. Once more it is the identity or a single odd transposition, a contradiction. □

For $n \geq 12$, two systems of two-point blocks cannot both have degree two. Lemmas 11 and 13 therefore cover every case where maximal nontrivial block systems on both original orbits have block size two.

Lemma 14 (One simultaneous swap with two large block pairs)**.** *Restore the string C-group and unique-crossing hypotheses. Suppose $m_1 = m_2 = 2$, $k_1, k_2 \geq 5$, and $\alpha = 1$. Then $r \leq \lfloor (n-1)/2 \rfloor$.*

Proof. Write $K = k_1 + k_2$. A failure forces $\beta = \gamma = 0$ and $r = K$. The set M consists of one involution τ, which exchanges both pairs of large blocks. Since it has exactly K point orbits, all of size two, it is fixed-point-free. It must be ρ_{i-1} or ρ_{i+1}; by reversal assume the former. Identify each pair of large blocks through τ. The full block image of G_i is the diagonal C_2, generated by τ.

Let $x = \rho_{i-2}$ if that index exists, and let C be the remaining distinguished generators of G_i other than τ, x. They commute with τ. Inside the permutations preserving both original orbits and inducing the stated diagonal block image, the centralizer of τ is

$$\mathrm{diag}(S_{k_1} \times S_{k_2}) \times \langle \tau \rangle.$$

The projection of $\langle C \rangle$ onto the diagonal factor is faithful: its kernel is contained in $\langle \tau \rangle$, whereas

IP gives $\langle C\rangle \cap \langle \tau\rangle = 1$. Write its image as H, and regard $\langle C\rangle$ as the graph of a character $\chi : H \to C_2$. Its marked family remains independent and has IP.

Except possibly $c = \rho_{i+1}$, every member of C fixes both crossing endpoints. Such a generator has $\chi = 0$ and fixes both corresponding row anchors. If x did not exist, then $|C| = K - 2$, and at least $K - 3$ members other than c would be independent in $S_{k_1-1} \times S_{k_2-1}$, whose bound is $K - 4$. This is impossible. Thus x exists and $|C| = K - 3$; the same bound shows that c exists as well. Whiston's equality clause now identifies

$$H_0 = \langle C \setminus \{c\}\rangle = S_{k_1-1} \times S_{k_2-1},$$

fixing exactly the two row anchors in the two original row sets.

The generator c cannot move both row anchors. If it did, every member of C other than c and its right neighbor would fix both anchors and their distinct c-images. This independent family has at least $K - 5$ members in $S_{k_1-2} \times S_{k_2-2}$, whose bound is $K - 6$. It cannot fix both anchors either, since then all $K - 3$ members of C would lie in the subgroup of independent length $K - 4$. After exchanging the two original orbits, it therefore joins exactly the first anchor to its other rows. A full point stabilizer together with a permutation moving the stabilized point generates the full symmetric group. Consequently

$$H = S_{k_1} \times S_{k_2-1},$$

with the second row anchor still fixed. Since χ vanishes on H_0, it vanishes on H: the point stabilizer S_{k_1-1} contains a transposition and normally generates S_{k_1}, and the second factor already belongs to H_0. Thus C acts diagonally, with no block exchanges.

The prefix of C at indices at most $i - 3$ is nonempty. Otherwise x would centralize all of C. It fixes both crossing endpoints and hence preserves all four large blocks setwise. On each block of the second orbit, centralizing the natural S_{k_2-1} fixes its unique fixed row anchor. Then τ, C, x preserve the two-point pair over that anchor, contradicting the transitivity of G_i on this original orbit. The suffix of C, starting at $i + 1$, is also nonempty.

These two interval subgroups commute and have trivial intersection by IP, so give a direct-product decomposition of H. The symmetric factors S_{k_1} and S_{k_2-1} are centerless and directly indecomposable; as in the accepted L047 factor lemma, they must be the two interval factors. The suffix contains c, which moves the first anchor, so it is S_{k_1}; the prefix is S_{k_2-1}. The total rank is their maximal independent length, so both markings attain their separate maxima.

Now x centralizes the entire diagonal S_{k_1}. Its action preserves each large block and hence is trivial on the first original orbit, because the centralizer of a natural symmetric group of degree at least three is trivial. On the second orbit it centralizes the endpoint omission of the prefix marking. For $k_2 \geq 6$, the accepted L047 extremal classification applies to S_{k_2-1}, including degrees five and six. In the ordinary Coxeter path the omission is the natural S_{k_2-2}; its centralizer on the k_2 rows is generated by the transposition of the omitted path endpoint with the permanently fixed row anchor. In the outer degree-six case its omission is transitive on those six rows. Its centralizer is trivial: a nontrivial centralizer of a transitive group is semiregular, so would contain a fixed-point-free element of order two or three. The centralizers of those cycle types in S_6 have orders $2^3 \cdot 3! = 48$ and $3^2 \cdot 2! = 18$, respectively; neither can contain this subgroup of order 120. Thus its centralizer on all seven rows is trivial as well.

When $k_2 = 5$, the direct classification of maximal S_4 strings in Lemma 12 shows that the omitted subgroup is either the natural S_3 or transitive D_8 on four of the five rows. In the latter case every centralizer fixes the unique fixed row anchor. Then x, just like C, fixes both points over that anchor, and τ only exchanges them. This contradicts the transitivity of G_i on the second original orbit. The S_3 case has the same two fixed rows and transposition centralizer as the ordinary case above.

In the ordinary case, on the large block containing the crossing endpoint, x fixes that endpoint and therefore cannot perform this transposition. On the other large block it may perform at

most that single transposition. Thus x is the identity or an odd transposition on all n points. Both are impossible. The outer case gives the identity immediately. This contradiction proves the rank bound. □

Scope and attribution. The two block systems and orbit-contraction strategy come from Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1, Section 5.1, physical pages 23–25. This formulation uses actual omission-orbit cuts for its forest and explicitly avoids assuming that a quotient marking has IP or that mere irredundance implies intransitive omissions. The complete L047 common theory and its older common normal-quotient/Whiston premises are required, including the intransitive equality lemma.

This bounded note proves the broad numerical reduction and the stated saturation consequences and uniform exclusions above, including the complete two-point-block case for $n \geq 12$ with primitive block images. It does not yet exclude the two-large-block parts of the residual families other than the one-swap branch with both block sizes at least five; nor does it prove the finer block drawings in Propositions 5.2–5.6, or complete Proposition 5.7. No new finite computation is required by these proofs, and no new formal theorem, original credit or conjecture claim is made.